

Genetic Counseling



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Objectives:

- Define Genetic Counselling
- Calculate the risk of genetic disorders
- Give examples of genetic counselling

How this session will work



Interactive case-based
discussion



You will calculate risks and
make decisions



Focus on real-life
counseling skills

Case 1

Couple (First cousins)

Child with spinal muscular atrophy

Planning their next pregnancy



Case 1 - Questions

- What is the inheritance pattern?
- Draw a pedigree
- What is the recurrence risk?
- How could you explain it?

Case 1 – Key Points

- Autosomal Recessive
- 25% recurrence risk
- Consanguinity increases risk
- Non-directive counseling

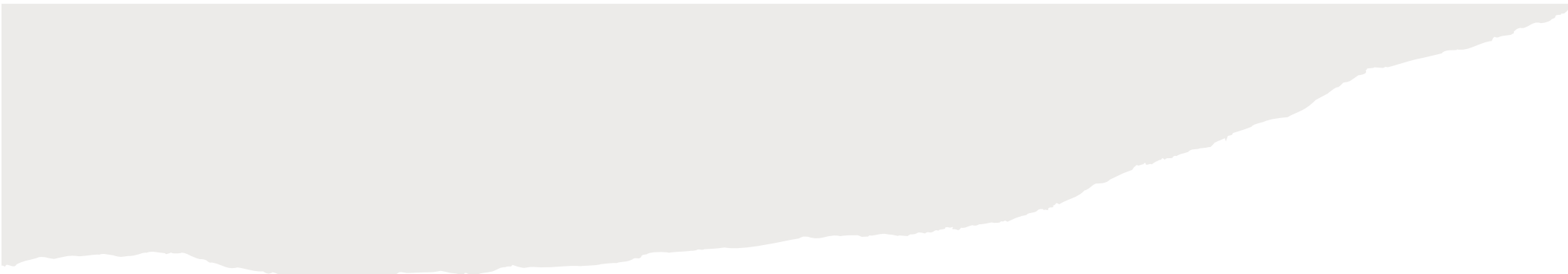
Case 1 – MCQ

- The couple had 3 unaffected children. What is the risk for the 4th child?
 - a) $< 25\%$
 - b) 25%
 - c) 100%
 - d) Zero

Case 2

- 35-year-old pregnant woman
- High risk non-invasive prenatal testing (NIPT) for Down syndrome





Your baby has Down syndrome.

You should consider termination



Case 2 - Questions

- Is NIPT diagnostic?
- Next step?
- How to explain risk?
- Counseling approach?

Case 2 – Key Points

- Screening vs diagnostic
- Confirm with amniocentesis
- Risk = probability, not certainty
- Use clear, empathic language

Case 3

- 28-year-old woman
- Family history of early breast cancer



Case 3 - Questions

- Inheritance pattern?
- Risk estimate?
- Should we test?
- Counseling considerations?

Case 3 – Key Points

- Autosomal dominant
- ~ 50% risk
- Pretest counseling essential
- Psychosocial impact

What is Genetic Counseling?



“The process of helping people understand and adapt to the medical, psychological and familial implications of genetic contributions to disease.”

This process integrates:

- Collecting & interpreting family & medical histories
- Education → Inheritance, testing, management, prevention, resources and research
- Addressing individual concerns relating to development & transmission of hereditary disorder
- Strong communication & support → promotes fully informed decision/adaptation

Indication for Genetic Counseling

Prenatal

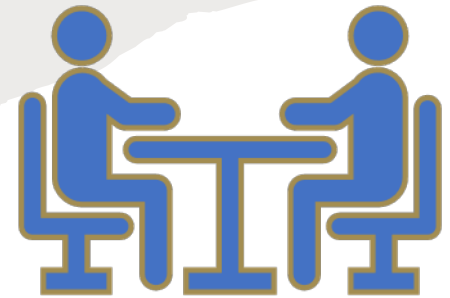
Pediatric

Adult

Key Skills

- Establish diagnosis
- Pedigree analysis
- Risk quantification
- Risk communication
- Non-directive, supportive and non-judgmental counseling

Communication and Support



- Communication is essential and a two-way process
- Setting → peaceful & comfortable; adequate time for discussion & questions
- Should:
 - Be a good listener
 - Be receptive to patients' fears & aspirations (expressed or unexpressed)
 - Present information in a clear & appropriate manner, with cultural sensitivity
 - Avoid technical terms or explain them carefully

- Language

ابنك متخلف عقليا

انت شايفه انو ولدك طبيعي!

بنتك غير الناس

طفل المنغولي

معاق

العيب منك

مافي امل

ايش الفايده انك تدخله مركز

ابنك عنده تشوهات خلقية

بنتك مراح تعيش مدة طويله

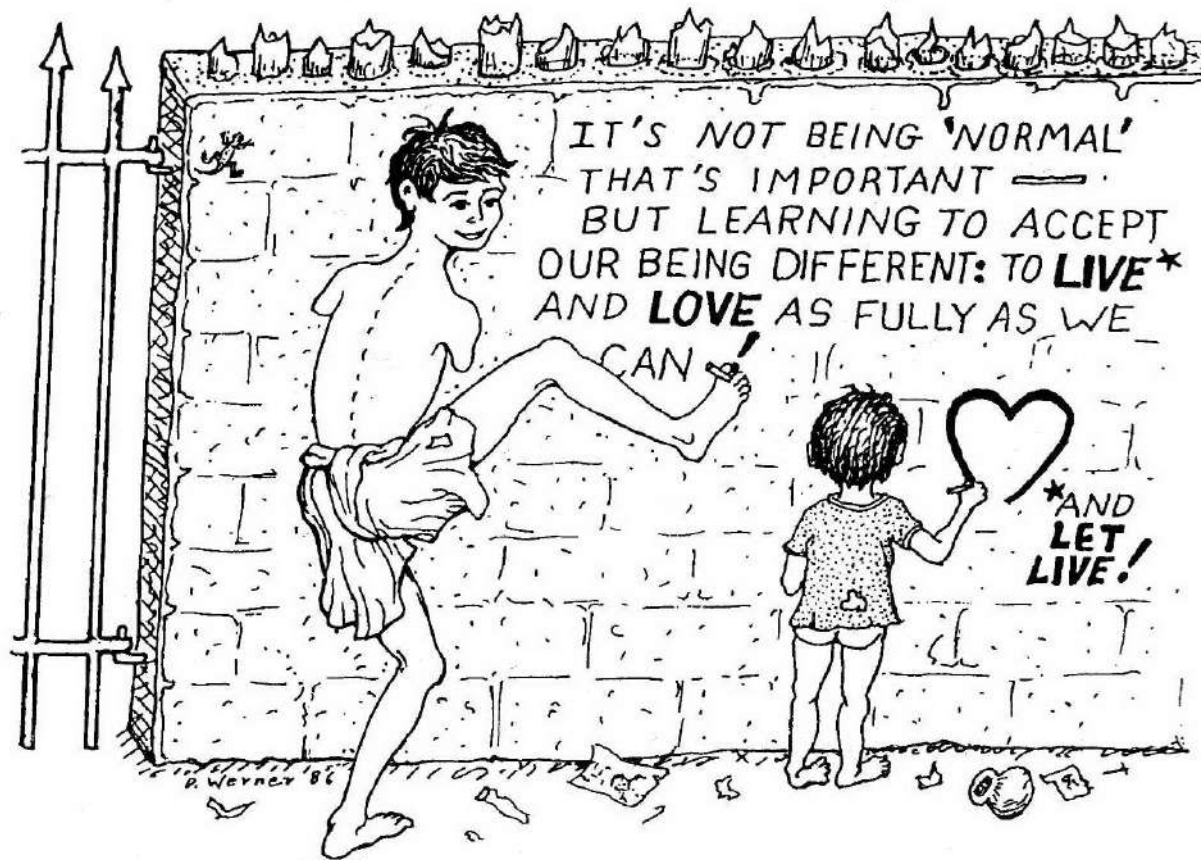
بتخسروا نفسكم على الفاضي

Take home message

- Genetic Counseling is a process
- Combine science + communication
- Support patient autonomy



DON'T FORGET



THANK YOU
Questions?